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Portfolio: Martin Ngoh

Martin Ngoh

Senior Data Scientist | AI Engineer

github.com/mngoh
linkedin.com/mngoh

Technical Skills

Languages **Proficient**: Python, PySpark, SQL, GIT, HTML
Experience: R, SAS, JavaScript, CSS

Tech Stack AWS, Azure, Tableau, Tensorflow, PyTorch, MLFlow, LangChain, numpy, pandas, matplotlib, seaborn, sklearn, Hugging Face, Prophet, OpenAI

Experience

Senior Data Scientist **July 2024 — Present**
Accenture Federal Services *Arlington, VA*

- Researched and engineered an automated document validation pipeline using AWS Textract APIs and LLM prompting to produce a readable JSON output that is compared to a person database. Reduces document validation by 100x and increases processing volume from 100 to 10,000+ documents in a single day.
- Developed and deployed a forecasting model using Prophet (MAPE: 12%), with self-healing, allowing for accurate prediction of computational load, resulting in more than $1 million in annual savings.
- Identified a novel approach to using AI for fraud detection. Independently devliered a tuned XGBoost (recall: 85%). Results were delivered to the client and future fraud detection work was assigned due to this initiative.
- Mentor junior data scientists, providing code reviews and coaching sessions to help build their confidence in delivering quality work while uplifting the team's performance as a whole.

Data Scientist **October 2021 — July 2024**
Deloitte Consulting *Arlington, VA*

- Strategized a road map with future AI initiatives for the client, helping secure a $30 million contract over 3 years.
- Boosted fraud detection by 8% by implementing Bayesian optimization (BayesSearchCV) to tune an ensemble model of neural network, elastic net and gradient boosting models.
- Refactored ML models to integrate with MLflow, enabling centralized model tracking and reducing documentation by 90%. This approach allowed for rapid experimentation, leading to $100 million more in yearly savings for the client.
- Automated a 12-hour program using Linux, reducing turnaround time from 3 days to 2 hours by leveraging cron jobs. Led a training program for 20 team members, which resulted in a 97% boost in efficiency.

Projects

GenAI RAG Model - ChatWithMe | ***Python, OpenAI, Whisper, Streamlit, NLTK*** **2026**

- Engineered an end-to-end RAG Pipeline to transform 10+ hours of raw voice recordings into a searchable personal knowledge base using Python and OpenAI's Whisper for transcription.
- Optimized data retrieval by chunking transcripts into a ChromaDB vector store; implemented vector embeddings, cosine similarity, and re-ranking to produce a context aware personalized chat bot.
- Tuned prompts by iteratively scoring model results and updating the prompt, leading to a an accurate RAG.

Education

M.S. Business Analytics, *Georgetown University (Varsity Men's Soccer Team)* 2022
B.S. Supply Chain Management and Analytics, *VCU (Varsity Men's Soccer Team)* 2020

Extracurricular

- Selected as a keynote speaker at Georgetown University's 2024 Black Alumni Summit; spoke on a panel of five AI experts discussing the implications of AI for the Black Community, to an audience of over 200.